



SCI_LAUNCH_A2_W02 · LAUNCH Science

Autumn 2 Week 2 teacher note

Objective (verbatim): Explain growth & stem cells.

Scheme of work: LAUNCH scheme of work, Autumn 2 week 2.

Pupil words: I am learning to explain how cells divide and then take on different jobs for growth and repair.

Reading gate: LAUNCH 11+; rough FK grade 6.3, age proxy ≈ 12.3 , 996 pupil-visible words excluding marked adult guidance. Formula is a rough screen, not an individual reading assessment.

Pearson foundation route: GCSE Biology 1BI0 Paper 1 and Combined Science 1SC0 Biology Paper 1, shared points 2.2 (repair), 2.5 (growth), 2.6 (differentiation), 2.8 (stem cells and meristems); shared Foundation. Official specification links: <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-biology-spec.pdf> and <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-combinedscience-spec.pdf>. The workbook's teaching approach is not a new specification point. No Biology-only B/HT point is assumed for Combined Foundation.

Gatsby 4: A plant nursery worker monitors meristem growth in young shoots; a biomedical scientist studies how tissues make replacement cells.

Sequence, success, retrieval

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40 minutes. Three one-minute resets are inside We do, Independent and Review; Arrival and resets make about eight minutes for settling and transitions.

Success at Starter, checked at Exit: I can distinguish making more cells from cells taking on a specialised job.; I can compare animal growth with plant growth.; I can identify a role for an animal stem cell or a plant meristem.

Retrieval from SoW: "last term" uses the Autumn 1 block in the supplied 2026/27 workbook. If a literal previous academic term is required, staff must check actual teaching rather than invent it.

When	Question	Answer	Exact SoW cell
last lesson	What two matching products follow body-cell mitosis and cell splitting?	Two daughter body cells with the same chromosome set.	LAUNCH scheme of work, Autumn 2 week 1
2-4 weeks	Which process moves particles from low to high concentration using energy?	Active transport.	LAUNCH scheme of work, Autumn 1 week 6 (Aut1 W6)
last term	In which part of a eukaryotic animal cell are chromosomes found?	The nucleus.	LAUNCH scheme of work, Autumn 1 week 1 (Aut1 W1)

Word meanings and examples (same in lesson and organiser):

Word	Tier	Definition	Example sentence
differentiation	3	a cell becoming specialised for a particular job	A developing skin cell differentiates to do a job in skin.



Word	Tier	Definition	Example sentence
stem cell	3	a cell that can divide and form specialised cell types	Some adult stem cells make replacement blood cells.
meristem	3	a plant region whose cells divide to make new cells	A root tip contains a meristem.
growth	2	an increase in size, including by making more cells	A shoot grows as its cells divide and elongate.

Faded example: I do full: In skin, mitosis adds cells; some stem-cell descendants differentiate for a tissue job. At a shoot tip, meristem cells divide, then elongate and differentiate. We do part-done: Skin: mitosis adds cells → differentiation gives a job. Shoot: meristem adds cells → lengthens cells → differentiation. Removed step: Elongation increases the length of plant cells during growth. Independent uses no filled answer. Fade next: Next time hide the animal differentiation arrow too; pupils explain both missing growth steps with the model.

Post-I-do show-me: Show me: does making more cells on its own mean those cells now have a specialised job? Expected: No; cell differentiation gives specialised roles. Every pupil shows a board, point or card. Re-model the named step if insecure.

F/M/H and named stretch

We do F support: Use two labelled branches (animal tissue and plant tip); pupils place the missing elongation card and point to the corresponding change.

We do M: Explain which animal and plant steps both make more cells and which give new jobs.

We do H stretch: Explain why a shoot can increase in length even after the division step has ended.

Independent F: Place divide, elongate and differentiate cards on the skin/shoot branches; omit elongate on the skin branch. Stem: In the shoot, first cells ...; then they ...

Independent M: Explain why mitosis alone does not give a new cell its specialised tissue job. Stem: More cells are made by ..., but a job develops when ...

Independent H: Compare growth in animal tissue with a plant shoot, including an appropriate stem-cell region and what each process adds. Stem: Both organisms ...; however, the plant also ... because ...

Go deeper: A tissue gains more cells but no new cell types. What does the model explain, and what is missing?

Named early-finisher stretch — Repair model audit: Correct a diagram labelling new skin cells “stem cell”; justify two changes. This is a harder correction/evaluation, not a repeated item.

On-screen interactions: Choose the missing label on each branch, then give a reason. uses native selects with a check/reset and aria-live feedback; We do 2 is a native 4-option all-pupil hinge without auto-grading. No storage. Original labelled diagram is simplified, not to scale. Its labels are checked in the fact ledger.

SoW practical status: no core practical is claimed. Prepared model pieces are an optional handling rehearsal with the pupil-facing Risk/Safe lines below.

AfL hinge and misconceptions

Question at We do 2: A gut-lining cell and a nerve cell share inherited DNA but have different structures. Which explanation fits this new pair?



Option	Correct?	Diagnosis
A. They differentiated into specialised cells with different jobs.	yes	Uses the new case accurately.
B. One cell must have lost every inherited gene.	no	Assumes specialisation requires losing all inherited genes.
C. Both cells must perform exactly the same job.	no	Confuses shared DNA with identical cell functions.
D. Neither cell can have developed through cell division.	no	Assumes differentiated cells could not have arisen after cell division.

Decision: About 3 in 4 or more correct: proceed; otherwise re-model a dividing stem cell and one specialised skin-cell branch with two paper cards, then ask the parallel hinge. No auto-grade. Parallel hinge: A muscle cell and a liver cell share inherited DNA. Must their jobs be identical? Answer: no; differentiation produces specialised cell types.

Misconception	How to spot	Correction
New cells must lose their DNA to specialise.	Pupil chooses B, claiming one cell lost every inherited gene.	Both branches keep DNA; differentiation means a different job, not loss of all DNA.
Cell division alone is sufficient to make specialised tissue.	Pupil chooses C, claiming the two specialised cells must have the same job.	Place a second card for differentiation; in plants include elongation as a further growth step.

Planned adaptations

Broad area	EEF five-a-day	Concrete action
Cognition & Learning	scaffolding	Scaffolding: Offer a two-branch, high-contrast strip; remove only one cue before asking the pupil to complete it.
Communication & Interaction	flexible grouping	Flexible grouping: Quiet pair rehearsal of the same comparison before any written route; point, sign or AAC accepted.
SEMH	cognitive/metacognitive strategies	Cognitive/metacognitive strategies: Privately preview the top comparison, agree a pause signal, then return to one card.
Sensory/Physical	technology	Technology: Offer large easy-grip model cards; speech-to-text or a photo of the labelled model can capture the response.
EAL (language, not SEN)	explicit instruction	Explicit instruction: Show the four science words with picture and example sentences; allow planning in any language. Keep M and H scientific demand available.

EAL language support remains available at M and H; the challenge stays open. Speak before write; the organiser's "My language" column is blank, without machine translation. Accept point, sign, speak, draw, type, write, exact scribing and staff-captured photo. Offer speech-to-text, TTS and immersive reader.

TA Focus and prompt ladder



At Independent choose at lesson start: teacher works with supported pupils and TA holds group, or TA supports and teacher holds group. Swap across the week. No public cold-calling is required; this is an invitation bank.

I do: "Watch the model. Point to what makes more cells, then what gives one cell a job."

We do: "One step is blank. Which part of the shoot becomes longer?"

Independent: "Which branch is your comparison about? Show the last step you know."

Least prompt first; no physical prompting: Wait five seconds; watch for an unaided card choice. → Point to the last completed branch or visual cue. → Ask "Which step makes more cells, and which gives a job?" → Model only the placement of the mitosis card; pupil places the next card. → Finish one branch together; pupil tries the other branch independently.

Exit handoff: adult states deepest prompt level, valid response mode and what pupil did independently; other adult chooses the next prompt.

Route	Invitation	Expected answer
F	What is the plant growth region called?	Meristem.
F	Which word means a cell gains a particular job?	Differentiation.
M	Why is division alone not enough for a specialised tissue?	New cells also need to differentiate.
M	What can also happen to plant cells as a shoot grows?	They elongate.
H	How are plant and animal growth alike and different?	Both involve division and differentiation; plant growth also includes cell elongation.
H	Why cannot one permanent stem-cell label describe every mature new skin cell?	Some daughter cells differentiate into specialised types.

Feedback, EHCP broad areas and regulation

Codes exactly: VF · WS · I · NS(+) · E · R · // · ? . VF at We do or Review; WS when supported with recorded prompt level; I when independent at Independent/Exit; NS(+) precise next step at Exit; // a real self-edit at Review; ? adult reads the changed words back on request; R after the pupil responds to NS(+) next lesson and staff judge, never automatic; E only for an actual curriculum observation.

What I record and where: note hinge response split and the named misconception on the staff quick-record grid; record actual code and prompt level in feedback. E: evidence captured on the school's evidence app. Never gather evidence during distress or store pupil data in HTML.

Pathways: high support, VF/WS regular. Build: WS/I mix. Grow: mostly independent, self-edit and response. Launch: independent self-assessment.

Regulation: Check in before the hinge, top rung and Independent. At the agreed pause signal give space and a single return step. Collect no evidence while a pupil is distressed. Re-enter at the last completed step, with one small choice and private correction.

lesson record poster: LAUNCH_SCI_A2_W02. Staff device or projector only; pupil phones not required.

Lundy Exit: pupil can say, point, sign, draw or write; teacher hears the answer and changes the next model/prompt. Four answers: goal at top, actual Review feedback, Exit NS(+), success criteria at Starter and Exit.



Sources, checks and limits

Every scientific and diagram label claim is listed in LESSON_SOURCE.json with a named official source and status. False distractors are deliberately incorrect and diagnosed. Read source claims in context; no medical claim about an individual. The animal/plant branches are simplified models. The classroom skin repair scene is illustrative; no individual treatment outcome is claimed.

Prepared model handling · practical safety

Kit: Large prepared division, elongation and differentiation cards, desk mat

Method: Place the prepared cards on the two labelled model branches.

Pupil Risk: Loose model cards could fall onto the floor.

Pupil Safe: Keep cards on the desk mat and pick up any that fall.

Staff confirm the local school risk assessment and exact kit; this does not assert a SoW core practical.

SLT planning and evaluation

Lesson code: SCI-LAUNCH-A2-W02

SoW reference: LAUNCH scheme of work, Autumn 2 week 2

Exact SoW outcome: Explain growth & stem cells.

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Evidence in this lesson: Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A2-W02. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Look at this week's seedling and skin-repair model. Point to one detail you notice. Staff: Offer a private, predictable start. Do not require previous knowledge.

Starter (3 min): Recall three ideas from earlier lessons. Show one answer at a time. Staff: All pupils respond; reveal the answers only after attempts. Revisit success criteria.

I Do (5 min): Watch the full model. Point to the step that answers today's question. Staff: Model one complete example; check every pupil's response to the show-me question.

We Do (7 min): Complete the missing step in the part-done model. Speak, then match the cards. Staff: Wait five seconds, gesture, then cue if needed. Support the route without supplying the missing answer.

We Do 2 (5 min): Choose one hinge answer. Show it at the same time as the group. Staff: Collect simultaneous responses, apply the decision rule and parallel hinge in the staff note.

Independent (8 min): A skin model shows division. What gives a new cell its job? Staff: At lesson start choose one adult with supported pupils and one holding the group; swap during the week. Record independence and prompt level.

Review (4 min): Tell a partner what you changed after feedback. Mark a real edit // and read it back when asked ?. Staff: Give a precise VF or NS(+) only when it actually happens. Invite a true self-edit and read-back.



Exit (3 min): Show what you know with a model and explain one reason. Staff: Hear the answer privately in a valid mode and use it to change the next example or prompt.

Planned adaptation — cognition learning: Scaffolding: Offer a two-branch, high-contrast strip; remove only one cue before asking the pupil to complete it.

Planned adaptation — communication interaction: Flexible grouping: Quiet pair rehearsal of the same comparison before any written route; point, sign or AAC accepted.

Planned adaptation — semh: Cognitive/metacognitive strategies: Privately preview the top comparison, agree a pause signal, then return to one card.

Planned adaptation — sensory physical: Technology: Offer large easy-grip model cards; speech-to-text or a photo of the labelled model can capture the response.

Planned adaptation — eal: Explicit instruction: Show the four science words with picture and example sentences; allow planning in any language. Keep M and H scientific demand available.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: flexible grouping; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: A skin model shows division. What gives a new cell its job? Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W03. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W03: Stem-cell ethics discussion (links RE/PSHE).

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Elongation increases the length of plant cells during growth. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "A gut-lining cell and a nerve cell share inherited DNA but have different structures. Which explanation fits this new pair?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Look at this week's seedling and skin-repair model. Point to one detail you notice. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: A skin model shows division. What gives a new cell its job? Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.



EAL: Model/shared rehearsal: use differentiation, stem cell, meristem, growth with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.